

exp_snipe_degree_frontier

Specialization: v1

TL;DR. The paper proves, to our knowledge, the first matched noiseless design-based minimax frontier for the finite-population total treatment effect simultaneously over the ℓ_1 coefficient-mass class and the uniformly bounded-outcome class, which is strictly larger when $B > 0$ and $d \geq 1$, under unit Bernoulli randomization and bounded-degree β -order neighborhood interference:

$$R_{n,\ell_1}^* \asymp_{\beta,p} R_{n,\infty}^* \asymp_{\beta,p} B^2 \min\left\{1, \frac{dA_d}{n}\right\} \asymp_{\beta,p} B^2 \min\left\{1, \frac{d\binom{d}{k_*}}{n}\right\}.$$

Clipped SNIPE, projected onto $[-2B, 2B]$ for the bounded-outcome class, attains the frontier, while unclipped SNIPE attains the linear branch. When $d \mid n$, complete directed d -blocks give canonical unclipped SNIPE exact worst-case MSE $B^2 A_d/m = B^2 d A_d/n$, so its global variance upper constant one is sharp. At $p = 1/2$, $A_d = 4 \sum_{1 \leq r \leq \min\{\beta, d\}, r \text{ odd}} \binom{d}{r}$, and $\beta = 1$ yields $B^2 \min\{1, d^2/n\}$. On complete blocks, the paper additionally solves the ℓ_1 block-local linear design-unbiased problem exactly, with finite minimax constant $B^2 A_d/m_t$ and a complete representer characterization.

1 Project justification

Gap. Under its $Y_{\max}$ coefficient-mass normalization, Theorem 1 of [1] gives a coarse $d^{\beta+2}/n$ -type stated worst-case SNIPE variance bound. Its Theorem 2 has a degree-free lower scale generated by Gaussian priors with unbounded coefficient mass and unbounded outcomes, so it is not a converse for either bounded class studied here; on the bounded classes the true linear-branch risk instead carries the factor $dA_d \asymp_{\beta,p} d\binom{d}{k_*}$, of order d^{k_*+1} when k_* is fixed. Thus the printed degree-free diagnosis does not describe bounded-outcome minimax difficulty. To our knowledge, the literature lacked an exact finite minimax constant and a complete optimal-weight characterization for block-local linear design-unbiased estimation on complete directed blocks; that structural question is resolved here rather than left open.

Niche. The local Bernoulli contrast representer has exact energy A_d , with $dA_d \asymp_{\beta,p} d\binom{d}{k_*}$, where k_* is the largest interaction order exposed by the all-one-versus-all-zero contrast. This separates within-neighborhood contrast complexity from the single additional out-degree overlap charge, works over both coefficient-mass-bounded and uniformly outcome-bounded schedules, and yields exact finite geometry on complete blocks.

Fill. The paper proves the simultaneous sandwich

$$c_{\beta,p} B^2 \min\left\{1, \frac{dA_d}{n}\right\} \leq R_{n,\ell_1}^* \leq R_{n,\infty}^* \leq C_{\beta,p} B^2 \min\left\{1, \frac{dA_d}{n}\right\},$$

and then converts it using the globally valid comparison $dA_d \asymp_{\beta,p} d\binom{d}{k_*}$. A single clipped SNIPE estimator attains the bounded-outcome upper bound, and the noiseless block mixtures, already contained in the ℓ_1 subclass, give the converse for both classes. Relative to the anchor's stated scales, the upper bound removes one degree factor, with additional gains when Bernoulli contrasts cancel, while the bounded-class converse restores the missing dA_d degree amplification. On complete blocks with $d \mid n$, canonical unclipped SNIPE has exact worst-case MSE $B^2 A_d/m$; the fair-coin specialization gives the exact odd-order energy identity and the $\beta = 1$ frontier $B^2 \min\{1, d^2/n\}$; and the separate ℓ_1 block-local theorem gives the exact finite minimax constant, the $\Psi_{b,t}$ risk formula, and the full asymptotic representer characterization.

2 Related work

Cortez-Rodriguez, Eichhorn, and Yu introduce SNIPE for low-order neighborhood interference under unit randomization [1]. Their Theorem 1 is the like-for-like ℓ_1 comparator because $Y_{\max} = \max_i \sum_{S'} |c_{i,S'}|$; the present variance

argument tightens their stated worst-case bound by charging overlap only once and replacing a coarse order count by the exact energy A_d . In the generic exposed-order regime this removes one full degree factor, with further gains when Bernoulli contrasts cancel. This is a comparison with their published upper bound, not a claim that SNIPE’s actual variance always attains that display. Their Theorem 2 gives a degree-free lower scale using Gaussian coefficient priors with unbounded support; it is not a bounded-class converse, and the paper does not claim to improve that theorem on its own unrestricted class. Rather, the new bounded-class least-favourable family shows that the bounded problem has linear-branch degree amplification $dA_d \asymp_{\beta,p} d\binom{d}{k_*}$, of order d^{k_*+1} , which the printed degree-free diagnosis does not capture. Harshaw, Sävje, and Wang give the general design-based Riesz framework [2], whereas the present paper derives the degree-sharp two-class frontier and the complete-block exact constant. Yu, Airolidi, Borgs, and Chayes provide an additive-interference benchmark under a different, unknown-network information set [3]; the fair-coin $\beta = 1$ theorem supplies the adjacent known-graph benchmark. Eichhorn, Khan, Ugander, and Yu use a variance bound to select among clusterings [4]. Their singleton-cluster corner $N = 1$, $C = d_{\max}$ is exactly the coarse unit-Bernoulli candidate sharpened here: this paper sharply recalibrates that specific candidate in their selection procedure, not their general clustered-design bound, and at $p = 1/2$, $\beta = 1$ the linear-regime consistency condition becomes $n \gg d^2$. Wang and Li develop covariate adjustment and conservative inference [5]; the unadjusted frontier and exact block-local benchmark here provide a scale against which such refinements can be judged.

3 Setup

Target (estimand / structural parameter): $\tau_n = n^{-1} \sum_{i \in V_n} [Y_i(\mathbf{1}) - Y_i(\mathbf{0})]$. Primary functional / estimator / diagnostic:

$$\hat{\tau}_n^{\text{up}} = \Pi_{[-B, B]} \left[n^{-1} \sum_{i \in V_n} Y_i^{\text{obs}} g_{i, \beta, p}(Z) \right]$$

- `n` — `population_size`; $\mathbb{N}$; `asymptotic_index`

Definition. Number of units.

- `V.n` — `unit_set`; $2^{\mathbb{N}}$; `population_index`

Definition. Finite population $V_n = \{1, \dots, n\}$.

- `G.n` — `directed_graph`; $V_n \times V_n$; `interference_graph`

Definition. Known directed interference graph, with self-loops allowed and counted in both degrees.

- `i` — `unit_index`; V_n ; `index`

Definition. Generic outcome-unit index.

- `j` — `unit_index`; V_n ; `index`

Definition. Generic assignment-coordinate index.

- `l` — `unit_index`; V_n ; `index`

Definition. Generic overlapping-neighborhood index.

- `N.i` — `in_neighborhood`; 2^{V_n} ; `interference_support`

Definition. Neighborhood $N_i = \{j \in V_n : (j, i) \in G_n\}$.

- `d` — `degree_bound`; $\{0, \dots, n\}$; `dependence_parameter`

Definition. Common upper bound on in-degrees and out-degrees.

- $\backslash\text{beta}$ — interaction_order; $\mathbb{N}$; model_complexity
Definition. Maximum polynomial interaction order.
- $\backslash\text{bar}\backslash\text{beta}_d$ — effective_interaction_order; $\{0, \dots, d\}$; frontier_index
Definition. Effective order $\bar{\beta}_d = \min\{\beta, d\}$.
- r — order_index; $\{1, \dots, \bar{\beta}_d\}$; index
Definition. Generic nonconstant interaction order when $d \geq 1$.
- p — Bernoulli_probability; $(0, 1)$; design_parameter
Definition. Known common treatment probability.
- Z — assignment_vector; $\{0, 1\}^{V_n}$; randomization
Definition. Realized assignment vector.
- P_Z — assignment_law; $2^{\{0, 1\}^{V_n}}$; design_law
Definition. Product Bernoulli assignment law of Z .
- z — assignment_argument; $\{0, 1\}^{V_n}$; function_argument
Definition. Generic assignment vector.
- S — neighborhood_subset; 2^{N_i} ; interaction_index
Definition. Generic subset of N_i .
- $\text{c}_{\{i, \text{S}\}}$ — coefficient; $\mathbb{R}$; causal_primitive
Definition. Coefficient of $\prod_{j \in S} z_j$ in unit i 's potential outcome.
- c — coefficient_schedule; $\mathbb{R}^{\{(i, S) : i \in V_n, S \subseteq N_i\}}$; causal_schedule
Definition. Collection of all coefficients $c_{i, S}$.
- B — coefficient_envelope; $[0, \infty)$; scale_parameter
Definition. Uniform ℓ_1 coefficient-mass envelope.
- $\text{Y}_i(\text{z})$ — potential_outcome; $\mathbb{R}$; causal_primitive
Definition. Potential outcome $Y_i(z) = \sum_{S \subseteq N_i} c_{i, S} \prod_{j \in S} z_j$.
- $\text{Y}_i^{\{\backslash\mathrm{obs}\}}$ — observed_outcome; $\mathbb{R}$; observed_data
Definition. Observed outcome $Y_i^{\text{obs}} = Y_i(Z)$.
- $\backslash\text{tau}_n$ — total_treatment_effect; $\mathbb{R}$; target_estimand
Definition. Finite-population contrast $\tau_n = n^{-1} \sum_{i \in V_n} [Y_i(\mathbf{1}) - Y_i(\mathbf{0})]$.
- $\backslash\text{Delta}_r(\text{p})$ — Bernoulli_contrast; $\mathbb{R}$; score_coefficient
Definition. Contrast $\Delta_r(p) = (1 - p)^r - (-p)^r$.
- $\text{k}_{\backslash\text{star}}$ — exposed_order; $\{0, \dots, \min\{\beta, d\}\}$; frontier_index
Definition. Exposed order constructed in `def : exposed - order`.
- $\backslash\text{mathcal G}_{\{n, d\}}$ — graph_class; $2^{V_n \times V_n}$; model_class
Definition. Bounded-degree directed graph class.
- $\backslash\text{mathcal C}_{\{n, d, \backslash\text{beta}\}}(\text{B})$ — coefficient_class; $\mathbb{R}^{\{(i, S) : i \in V_n, S \subseteq N_i\}}$; model_class
Definition. Bounded-mass low-order coefficient schedules.

- $\mathcal{M}_{n,d,\beta}(B)$ — `finite_population_model_class`; $\mathcal{G}_{n,d} \times \mathcal{C}_{n,d,\beta}(B)$; `model_class`
Definition. Graph-and-schedule class for minimax risk.
- $g_{i,\beta,p}$ — `SNIPE_score`; $\{0,1\}^{V_n} \rightarrow \mathbb{R}$; `estimator_weight`
Definition. Centered Bernoulli contrast representer constructed in `def : snipe - score`.
- $\widehat{\tau}_n(\mathrm{SNIPE})$ — `design_based_estimator`; $\mathbb{R}$; `estimator`
Definition. Unclipped SNIPE estimator.
- $\widehat{\tau}_n(\mathrm{up})$ — `clipped_design_based_estimator`; $\mathbb{R}$; `attaining_estimator`
Definition. Projection of SNIPE onto $[-B, B]$.
- $\widehat{\tau}_n$ — `generic_estimator`; $\mathbb{R}$; `minimax_competitor`
Definition. Estimator measurable in observed data and the graph.
- $\mathbb{E}Z$ — `design_expectation`; $\mathbb{R} \rightarrow \mathbb{R}$; `design_expectation`
Definition. Expectation over assignment, with graph and potential outcomes fixed.
- $R_n^*(d,\beta,p,B)$ — `minimax_MSE`; $[0,\infty)$; `frontier`
Definition. Design-based minimax mean squared error constructed in `def : minimax - risk`.
- m — `block_count`; $\{0,\dots,n\}$; `construction_index`
Definition. Active block count $m = \lfloor n/d \rfloor$ for $d \geq 1$.
- q — `active_unit_count`; $\{0,\dots,n\}$; `construction_index`
Definition. Number of active units $q = md$.
- ρ — `active_population_share`; $(0,1]$; `construction_parameter`
Definition. Active population share $\rho = q/n$ for $d \geq 1$.
- b — `block_index`; $\{1,\dots,m\}$; `index`
Definition. Generic active-block index.
- V_b — `block`; 2^{V_n} ; `least_favorable_component`
Definition. Active complete directed block of size d .
- $\mathcal{G}_{n,d}^{\mathrm{blk}}$ — `block_graph`; $\mathcal{G}_{n,d}$; `least_favorable_graph`
Definition. Graph with complete directed active blocks, including loops, and isolated inactive units.
- g_d — `block_score`; $\{0,1\}^{\{1,\dots,d\}} \rightarrow \mathbb{R}$; `least_favorable_direction`
Definition. Block centered contrast score constructed in `def : block - score - energy`.
- A_d — `block_score_energy`; $[0,\infty)$; `combinatorial_energy`
Definition. Block score energy constructed in `def : block - score - energy`.
- A_i — `local_score_energy`; $[0,\infty)$; `combinatorial_energy`
Definition. Exact local score energy at outcome unit i .
- h_d — `normalized_representer`; $\{0,1\}^{\{1,\dots,d\}} \rightarrow \mathbb{R}$; `least_favorable_direction`
Definition. Normalized block representer for $d \geq 1$.

- T — `block_subset`; $2^{\{1, \dots, d\}}$; `interaction_index`

Definition. Generic subset of block coordinates.

- $h_{\{d, T\}}$ — `raw_monomial_coefficient`; $\mathbb{R}$; `construction_coefficient`

Definition. Raw-monomial coefficient in $h_d(z) = \sum_T h_{d, T} \prod_{j \in T} z_j$.

- $H_{\{\beta, p\}}$ — `representer_mass_constant`; $(0, \infty)$; `construction_constant`

Definition. Uniform raw-monomial coefficient-mass bound for h_d .

- s — `baseline_halfwidth`; $(0, \infty)$; `construction_parameter`

Definition. Baseline halfwidth $s = B/2$ when $B > 0$.

- f_s — `baseline_density`; $\mathbb{R} \rightarrow [0, \infty)$; `prior_density`

Definition. Cosine-squared density on $[-s, s]$.

- U_b — `block_baseline`; $[-s, s]$; `least_favorable_nuisance`

Definition. Prior-randomized fixed constant coefficient for active block b .

- σ — `tilt_sign`; $\{-1, 1\}$; `least_favorable_index`

Definition. Index of the two prior mixtures.

- δ — `tilt_amplitude`; $(0, \infty)$; `separation_parameter`

Definition. Score-aligned coefficient tilt constructed in `def : block - family`.

- Π_{σ} — `prior_on_schedules`; $2^{\mathcal{M}_{n, d, \beta}(B)}$; `least_favorable_family`

Definition. Two fuzzy prior mixtures over fixed noiseless schedules.

- P_{σ} — `prior_predictive_law`; $2^{\{0, 1\}^{V_n} \times \mathbb{R}^n}$; `lower_bound_law`

Definition. Law of assignment and observed outcomes after integrating Π_{σ} .

- $c_{\{\beta, p\}}$ — `lower_constant`; $(0, \infty)$; `theorem_constant`

Definition. Positive constant depending only on fixed β and p .

- $C_{\{\beta, p\}}$ — `upper_constant`; $(0, \infty)$; `theorem_constant`

Definition. Finite constant depending only on fixed β and p .

- $\mathcal{P}_d$ — `polynomial_space`; $L^2(P_Z)$; `Riesz_space`

Definition. Span of raw monomials $\prod_{j \in S} z_j$ with $|S| \leq \bar{\beta}_d$.

- L_d — `contrast_functional`; $\mathcal{P}_d \rightarrow \mathbb{R}$; `Riesz_functional`

Definition. All-one versus all-zero functional $L_d(f) = f(\mathbf{1}_d) - f(\mathbf{0}_d)$.

- h — `candidate_perturbation`; $\mathcal{P}_d$; `optimization_variable`

Definition. Candidate normalized contrast perturbation.

- f — `candidate_polynomial`; $\mathcal{P}_d$; `optimization_variable`

Definition. Generic polynomial to which the block contrast functional is applied.

- $\text{PerturbProg}_{\{d, \beta, p\}}$ — `Riesz_optimization`; $\mathbb{R}$; `lower_bound_handle`

Definition. Normalized perturbation program constructed in `def : perturbation - program`.

- $\text{WeightProg}_{\{d, \beta, p\}}$ — `Riesz_optimization`; $\mathbb{R}$; `estimator_handle`

Definition. Unbiased score-weight program constructed in `def : weight - program`.

- `w` — candidate_weight; $L^2(P_Z)$; optimization_variable

Definition. Candidate block-local weight.

- `n.t` — sequence_population_size; $\mathbb{N}$; sequence_population_index

Definition. Population size along the exact-constant sequence.

- `V_{n.t}` — sequence_unit_set; $2^{\mathbb{N}}$; sequence_unit_set

Definition. Finite population along the sequence, $V_{n_t} = \{1, \dots, n_t\}$.

- `d.t` — sequence_degree; $\mathbb{N}$; sequence_degree_index

Definition. Block size along the exact-constant sequence.

- `G.t` — sequence_block_graph; $\mathcal{G}_{n_t, d_t}$; fixed_complete_block_graph

Definition. Disjoint union of complete directed d_t -blocks with loops.

- `N_i^{G.t}` — sequence_in_neighborhood; $2^{V_{n_t}}$; fixed_graph_neighborhood

Definition. In-neighborhood of i in the fixed graph G_t .

- `b(i)` — block_membership_map; $\{1, \dots, n_t/d_t\}$; block_membership_index

Definition. Block containing i in G_t .

- `Z_{b(i)}` — block_assignment_subvector; $\{0, 1\}^{V_{b(i)}}$; block_assignment

Definition. Restriction of the assignment vector to the block containing i in G_t .

- `w_{i,t}` — sequence_weight; $L^2(P_Z)$; block_local_weight

Definition. Assignment weight local to the block containing i .

- `\widehat{\tau}_{w,t}` — restricted_estimator; $\mathbb{R}$; block_local_linear_estimator

Definition. Block-local linear estimator generated by weights $w_{i,t}$.

- `\mathcal{E}_t^{\{\mathrm{loc}, \mathrm{lin}\}}` — restricted_estimator_class; $2^{\mathbb{R}}$; proved_restricted_estimator_class

Definition. Block-local linear design-unbiased estimator class constructed in `def : local – linear – class`.

- `\mathcal{M}_t(G.t)` — fixed_graph_schedule_class; $2^{\mathbb{R}^{\{(i,S): i \in V_{n_t}, S \subseteq N_i^{G_t}\}}}$; fixed_graph_schedule_class

Definition. Admissible coefficient schedules on the fixed graph G_t .

- `R_t^{\{\mathrm{loc}, \mathrm{lin}\}}` — restricted_minimax_risk; $[0, \infty)$; exact_restricted_minimax_risk

Definition. Restricted minimax risk constructed in `def : local – linear – class`.

- `\mathcal{C}_{n,d,\beta}^{\infty}(B)` — bounded_outcome_coefficient_class; $\mathbb{R}^{\{(i,S): i \in V_n, S \subseteq N_i\}}$; model_class

Definition. Low-order coefficient schedules whose induced potential outcomes are uniformly bounded in absolute value by B .

- `\mathcal{M}_{n,d,\beta}^{\infty}(B)` — bounded_outcome_finite_population_model_class; $\mathcal{G}_{n,d} \times \mathcal{C}_{n,d,\beta}^{\infty}(B)$; model_class

Definition. Bounded-degree, low-order, uniformly bounded-outcome graph-and-schedule class.

- `R_{n,\ell_1}^{\star}(d,\beta,p,B)` — coefficient_mass_minimax_MSE; $[0, \infty)$; frontier

Definition. Design-based minimax MSE over the ℓ_1 coefficient-mass class; it equals the previously defined $R_n^{\star}(d, \beta, p, B)$.

- $R_{\{n, \infty\}^{\star}(d, \beta, p, B)}$ — `bounded_outcome_minimax_MSE`; $[0, \infty)$; `frontier`

Definition. Design-based minimax MSE over the uniformly bounded-outcome model class.

- $\widehat{\tau}_{\{n, \infty\}^{\uparrow}}$ — `bounded_outcome_clipped_design_based_estimator`; $\mathbb{R}$; `attaining_estimator`

Definition. Projection of SNIPE onto $[-2B, 2B]$.

4 Assumptions

Assumption 1 (`ass:bernoulli-design`). Z_j are mutually independent and $\Pr(Z_j = 1) = p$ for every $j \in V_n$.

Standard (*unit-level Bernoulli randomization*, [1]).

Assumption 2 (`ass:bounded-degree`). Self-loops are allowed and count in both degrees, with $\max_{i \in V_n} |N_i| \leq d$ and $\max_{j \in V_n} |\{i \in V_n : j \in N_i\}| \leq d$.

Standard (*bounded in- and out-degree neighborhood interference with self-loops*, [1]).

Assumption 3 (`ass:low-order`). $c_{i,S} = 0$ whenever $|S| > \beta$.

Standard (*bounded polynomial interaction order*, [1]).

Assumption 4 (`ass:bounded-coefficient-mass`). $\sum_{S \subseteq N_i} |c_{i,S}| \leq B$ for every $i \in V_n$.

Standard (*SNIPE $Y_{\max}$ coefficient-mass normalization*, [1]).

5 Definitions

Definition 1 (`def:exposed-order`). Defined object. *effective and exposed orders*

Construction. Set $\bar{\beta}_d := \min\{\beta, d\}$ for every $d \geq 0$; in particular $\bar{\beta}_0 = 0$. If $\bar{\beta}_d \geq 1$, set $k_{\star}(d, \beta, p) := \max\{r \in \{1, \dots, \bar{\beta}_d\} : \Delta_r(p) \neq 0\}$; since $\Delta_1(p) = 1$, this index set contains $r = 1$, so its maximum is well defined. When $\bar{\beta}_d = 0$, equivalently $d = 0$ or $\beta = 0$, set $k_{\star}(d, \beta, p) := 0$ as a totality convention only, never as an attained interaction order. Thus the maximum case applies exactly when both $d \geq 1$ and $\beta \geq 1$; every statement below fixes $\beta \geq 1$.

Definition 2 (`def:zero-degree-conventions`). Defined object. *zero-degree conventions*

Construction. Set $\binom{0}{0} := 1$, $g_0 := 0$, and $A_0 := 0$. Interpret sums over nonempty subsets of $\{1, \dots, 0\}$ as zero; its assignment space has one empty vector and the empty product equals one. Set $h_0 := 0$ only as an empty-object convention, and never interpret it as g_0/A_0 .

Definition 3 (`def:graph-class`). Defined object. *bounded-degree graph class*

Construction. $\mathcal{G}_{n,d} := \{G_n \in 2^{V_n \times V_n} : G_n \text{ satisfies } \text{ass} : \text{bounded} - \text{degree}\}$.

Member properties. *ass:bounded-degree*.

Definition 4 (`def:coefficient-class`). Defined object. *bounded low-order coefficient class*

Construction.

$\mathcal{C}_{n,d,\beta}(B) := \{c : c \text{ satisfies } \text{ass} : \text{low} - \text{order} \text{ and } \text{ass} : \text{bounded} - \text{coefficient} - \text{mass}\}$

Member properties. *ass:low-order; ass:bounded-coefficient-mass*.

Definition 5 (`def:model-class`). Defined object. *bounded-degree SNIPE model class*

Construction. $\mathcal{M}_{n,d,\beta}(B) := \{(G_n, c) : G_n \in \mathcal{G}_{n,d}, c \in \mathcal{C}_{n,d,\beta}(B)\}$, where $\mathcal{G}_{n,d}$ and $\mathcal{C}_{n,d,\beta}(B)$ are the classes in `def : graph - class` and `def : coefficient - class`, respectively.

Member properties. *ass:bounded-degree; ass:low-order; ass:bounded-coefficient-mass*.

Definition 6 (def:snipe-score). Defined object. *centered SNIPE score*
Construction.

$$g_{i,\beta,p}(z) := \sum_{r=1}^{\bar{\beta}_d} \frac{\Delta_r(p)}{[p(1-p)]^r} \sum_{\substack{S \subseteq N_i \\ |S|=r}} \prod_{j \in S} (z_j - p)$$

Definition 7 (def:snipe-estimator). Defined object. *SNIPE and clipped SNIPE estimators*

Construction. $\hat{\tau}_n^{\text{SNIPE}} := n^{-1} \sum_{i \in V_n} Y_i^{\text{obs}} g_{i,\beta,p}(Z)$ and $\hat{\tau}_n^{\text{up}} := \Pi_{[-B,B]}(\hat{\tau}_n^{\text{SNIPE}})$, where $\Pi_{[-B,B]}$ is Euclidean projection.

Definition 8 (def:minimax-risk). Defined object. *design-based minimax risk*

Construction.

$$R_n^*(d, \beta, p, B) := \inf_{\hat{\tau}_n} \sup_{(G_n, c) \in \mathcal{M}_{n,d,\beta}(B)} \mathbb{E}_Z[(\hat{\tau}_n - \tau_n)^2]$$

, where $\hat{\tau}_n$ is measurable in $(G_n, Z, \{Y_i^{\text{obs}} : i \in V_n\})$.

Definition 9 (def:block-score-energy). Defined object. *block score, energy, and raw expansion*

Construction. For $d \geq 1$ and a d -coordinate Bernoulli vector, set

$$g_d(z) := \sum_{r=1}^{\bar{\beta}_d} \frac{\Delta_r(p)}{[p(1-p)]^r} \sum_{\substack{S \subseteq \{1, \dots, d\} \\ |S|=r}} \prod_{j \in S} (z_j - p)$$

, $A_d := \sum_{r=1}^{\bar{\beta}_d} \binom{d}{r} \Delta_r(p)^2 / [p(1-p)]^r$, and $h_d := g_d / A_d = \sum_{T \subseteq \{1, \dots, d\}} h_{d,T} \prod_{j \in T} z_j$, the last equality being the raw-monomial expansion.

Definition 10 (def:block-family). Defined object. *continuous-baseline score-aligned block family*

Construction. For $B > 0$ and $1 \leq d \leq n$, set $m := \lfloor n/d \rfloor$, $q := md$, and $\rho := q/n$. Partition q units into m blocks V_b of size d ; include every within-block directed arrow, including loops, and leave the remaining $n - q$ units isolated. Set $s := B/2$, $f_s(u) := s^{-1} \cos^2(\pi u / (2s)) \mathbf{1}\{|u| \leq s\}$, and draw $U_1, \dots, U_m$ independently from f_s . Let $H_{\beta,p} := \sup_{d \geq 1} \sum_T |h_{d,T}|$ and choose $\delta := B \min\{(2H_{\beta,p})^{-1}, (4\pi)^{-1}\} \min\{1, \sqrt{A_d/m}\}$. Under Π_σ , active $i \in V_b$ has $c_{i,T} := U_b \mathbf{1}\{\bar{T} = \emptyset\} + \sigma \delta h_{d,T}$ for $|T| \leq k_*$, and all other coefficients, including those of inactive units, are zero. The U_b randomize fixed potential-outcome schedules only; they are not observation noise.

Definition 11 (def:perturbation-program). Defined object. *normalized perturbation Riesz program*

Construction. For $d \geq 1$, $\text{PerturbProg}_{d,\beta,p} := \min_{h \in \mathcal{P}_d} \mathbb{E}_Z[h(Z)^2]$ subject to $L_d(h) = 1$.

Definition 12 (def:weight-program). Defined object. *unbiased score-weight Riesz program*

Construction. For $d \geq 1$, $\text{WeightProg}_{d,\beta,p} := \min_{w \in L^2(P_Z)} \mathbb{E}_Z[w(Z)^2]$ subject to $\mathbb{E}_Z[w(Z)] = 0$ and $\mathbb{E}_Z[w(Z) \prod_{j \in S} Z_j] = 1$ for every nonempty $S \subseteq \{1, \dots, d\}$ with $|S| \leq \bar{\beta}_d$.

Definition 13 (def:local-linear-class). Defined object. *block-local linear unbiased coefficient class*

Construction. On the fixed graph G_t , set

$$\mathcal{M}_t(G_t) := \{c : c_{i,S} = 0 \text{ unless } S \subseteq N_i^{G_t} \text{ and } |S| \leq \bar{\beta}_{d_t}, \sum_{S \subseteq N_i^{G_t}} |c_{i,S}| \leq B \text{ for every } i \in V_{n_t}\}$$

Member properties. *ass:low-order; ass:bounded-coefficient-mass.*

Definition 14 (def:local-linear-estimator-class). Defined object. *block-local linear unbiased estimator class*

Construction. *Set*

$$\mathcal{E}_t^{\text{loc,lin}} := \{\hat{\tau}_{w,t} = n_t^{-1} \sum_{i \in V_{n_t}} w_{i,t}(Z_{b(i)})Y_i(Z) : w_{i,t} \in L^2(P_Z) \text{ is measurable with respect to } Z_{b(i)}, \mathbb{E}_Z[w_{i,t}] = 0, \text{ and } \mathbb{E}_Z[w_{i,t} \prod_{j \in S} Z_j] = 1 \text{ for every nonempty } S \subseteq N_i^{G_t} \text{ with } |S| \leq \bar{\beta}_{d_t}\}$$

Definition 15 (def:local-linear-risk). Defined object. *fixed-graph block-local linear minimax risk*

Construction. *Define*

$$R_t^{\text{loc,lin}} := \inf_{\hat{\tau}_{w,t} \in \mathcal{E}_t^{\text{loc,lin}}} \sup_{c \in \mathcal{M}_t(G_t)} \mathbb{E}_Z[(\hat{\tau}_{w,t} - \tau_{n_t})^2]$$

; the graph is fixed and is not varied in this supremum.

Definition 16 (def:bounded-outcome-coefficient-class). Defined object. *uniformly bounded-outcome low-order coefficient class*

Construction.

$$\mathcal{C}_{n,d,\beta}^\infty(B) := \{c : c_{i,S} = 0 \text{ whenever } |S| > \beta, \sup_{z \in \{0,1\}^{V_n}} |Y_i(z)| \leq B \text{ for every } i \in V_n\}$$

Member properties. *ass:low-order.*

Definition 17 (def:bounded-outcome-model-class). Defined object. *bounded-degree uniformly bounded-outcome model class*

Construction.

$$\mathcal{M}_{n,d,\beta}^\infty(B) := \{(G_n, c) : G_n \in \mathcal{G}_{n,d}, c \in \mathcal{C}_{n,d,\beta}^\infty(B)\}$$

, where the two component classes are those in def : graph – class and def : bounded – outcome – coefficient – class.

Member properties. *ass:bounded-degree; ass:low-order.*

Definition 18 (def:two-class-minimax-risks). Defined object. *coefficient-mass and bounded-outcome minimax risks*

Construction. $R_{n,\ell_1}^\star(d, \beta, p, B) := R_n^\star(d, \beta, p, B)$ and

$$R_{n,\infty}^\star(d, \beta, p, B) := \inf_{\hat{\tau}_n} \sup_{(G_n, c) \in \mathcal{M}_{n,d,\beta}^\infty(B)} \mathbb{E}_Z[(\hat{\tau}_n - \tau_n)^2]$$

, where $\hat{\tau}_n$ is measurable in $(G_n, Z, \{Y_i^{\text{obs}} : i \in V_n\})$.

Inputs. *def:minimax-risk; def:bounded-outcome-model-class.*

Definition 19 (def:bounded-outcome-clipped-snipe). Defined object. *bounded-outcome clipped SNIPE estimator*

Construction. $\hat{\tau}_{n,\infty}^{\text{up}} := \Pi_{[-2B, 2B]}(\hat{\tau}_n^{\text{SNIPE}})$, where $\Pi_{[-2B, 2B]}$ is Euclidean projection.

Inputs. *def:snipe-estimator.*

6 Main results

Lemma 1 (lem:block-energy-representer). *For fixed $\beta \geq 1$, $p \in (0, 1)$, and $d \geq 1$, $A_d \asymp_{\beta, p} \binom{d}{k_\star}$, $\mathbb{E}_Z[g_d(Z)f(Z)] = L_d(f)$ for every $f \in \mathcal{P}_d$, $h_d(\mathbf{1}_d) - h_d(\mathbf{0}_d) = 1$, and $\mathbb{E}_Z[h_d(Z)^2] = A_d^{-1}$. The unique minimizers of $\text{def} : \text{perturbation} - \text{program}$ and $\text{def} : \text{weight} - \text{program}$ are respectively $h_d = g_d/A_d$, with value A_d^{-1} , and $w = g_d$, with value A_d . Moreover $H_{\beta, p} = \sup_{d \geq 1} \sum_T |h_{d, T}| < \infty$.*

Proof. Put $v := p(1 - p)$, $X_j := Z_j - p$, $X_S := \prod_{j \in S} X_j$, and $Z_T := \prod_{j \in T} Z_j$. Independence and centering under $\text{ass} : \text{bernoulli} - \text{design}$ give

$$\mathbb{E}_Z[X_S X_T] = \mathbf{1}\{S = T\} v^{|S|}.$$

Moreover, for $S \neq \emptyset$,

$$\mathbb{E}_Z[X_S Z_T] = \mathbf{1}\{S \subseteq T\} v^{|S|} p^{|T| - |S|}.$$

Thus, if $1 \leq t := |T| \leq \bar{\beta}_d$, then

$$\begin{aligned} \mathbb{E}_Z[g_d(Z) Z_T] &= \sum_{r=1}^t \binom{t}{r} \Delta_r(p) p^{t-r} \\ &= \sum_{r=1}^t \binom{t}{r} (1-p)^r p^{t-r} - \sum_{r=1}^t \binom{t}{r} (-p)^r p^{t-r} \\ &= \{(p + 1 - p)^t - p^t\} - \{(p - p)^t - p^t\} = 1. \end{aligned}$$

For $T = \emptyset$, the expectation is zero. Since the raw monomials of degree at most $\bar{\beta}_d$ span $\mathcal{P}_d$, linearity proves

$$\mathbb{E}_Z[g_d(Z)f(Z)] = L_d(f), \quad f \in \mathcal{P}_d.$$

Orthogonality of the centered monomials also gives

$$\mathbb{E}_Z[g_d(Z)^2] = \sum_{r=1}^{\bar{\beta}_d} \binom{d}{r} \frac{\Delta_r(p)^2}{v^r} = A_d.$$

Let $k := k_\star(d, \beta, p)$. By $\text{def} : \text{exposed} - \text{order}$, every term above order k vanishes, whereas the order- k coefficient is positive. For fixed β and p , the finitely many nonzero coefficients $\Delta_r(p)^2/v^r$ are bounded above and away from zero. Also

$$\sum_{r=1}^k \binom{d}{r} \leq C_\beta \binom{d}{k}.$$

Indeed, this follows from monotonicity of the binomial coefficients through order β when $d \geq 2\beta$, and the finitely many remaining values of d are absorbed into C_β . The order- k summand supplies the reverse inequality. Hence

$$A_d \asymp_{\beta, p} \binom{d}{k_\star}.$$

For any feasible h in $\text{def} : \text{perturbation} - \text{program}$, the representer identity and Cauchy–Schwarz imply

$$1 = L_d(h)^2 = \langle g_d, h \rangle_{L^2(P_Z)}^2 \leq A_d \mathbb{E}_Z[h(Z)^2].$$

Equality holds only when h is proportional to g_d , and the constraint fixes the proportionality constant as A_d^{-1} . Therefore the unique minimizer is $h_d = g_d/A_d$, its value is A_d^{-1} , and

$$h_d(\mathbf{1}_d) - h_d(\mathbf{0}_d) = L_d(h_d) = 1, \quad \mathbb{E}_Z[h_d(Z)^2] = A_d^{-1}.$$

If w is feasible in $\text{def} : \text{weight} - \text{program}$, its moment restrictions and linearity give

$$\langle w, f \rangle_{L^2(P_Z)} = L_d(f) = \langle g_d, f \rangle_{L^2(P_Z)}, \quad f \in \mathcal{P}_d.$$

Consequently $w - g_d$ is orthogonal to $\mathcal{P}_d$. Since $g_d \in \mathcal{P}_d$,

$$\mathbb{E}_Z[w(Z)^2] = A_d + \mathbb{E}_Z[(w(Z) - g_d(Z))^2] \geq A_d,$$

with equality only for $w = g_d$. This proves the second unique-minimizer assertion.

Finally,

$$X_S = \sum_{T \subseteq S} (-p)^{|S| - |T|} Z_T.$$

The triangle inequality for raw coefficients therefore yields

$$\sum_T |h_{d,T}| \leq \frac{1}{A_d} \sum_{r=1}^{\bar{\beta}_d} \binom{d}{r} \frac{|\Delta_r(p)|}{v^r} (1+p)^r.$$

Only orders at most k_* with nonzero contrast contribute. The same binomial comparison used for A_d , together with the positive order- k_* term in its denominator, bounds the last display uniformly over $d \geq 1$ by a finite constant depending only on β, p . Thus $H_{\beta,p} < \infty$. $\square$

Justification. The exact score energy and the two minimum-norm identities identify the sharp scale and a valid bounded-mass lower-bound direction. *Gap.* The score construction in [1] does not isolate this normalized contrast representer, its exact energy, or the two optimization programs. *Consumer.* It supplies the SNIPE variance scale and the admissible direction for the noiseless fuzzy converse.

Lemma 2 (lem:overlap-count). *For every $r \geq 1$, including $d = 0$, and every $i \in V_n$,*

$$\sum_{l \in V_n} \binom{|N_i \cap N_l|}{r} = \sum_{\substack{S \subseteq N_i \\ |S|=r}} \#\{l \in V_n : S \subseteq N_l\} \leq d \binom{|N_i|}{r} \leq d \binom{d}{r}.$$

Proof. Double-count the pairs (l, S) such that $l \in V_n$, $S \subseteq N_i \cap N_l$, and $|S| = r$. Counting first by l and then by S gives

$$\sum_{l \in V_n} \binom{|N_i \cap N_l|}{r} = \sum_{\substack{S \subseteq N_i \\ |S|=r}} \#\{l \in V_n : S \subseteq N_l\}.$$

For each nonempty S , choose $j_S \in S$. The out-degree part of $\text{ass} : \text{bounded} - \text{degree}$ implies

$$\#\{l : S \subseteq N_l\} \leq \#\{l : j_S \in N_l\} \leq d.$$

Summing this inequality over the $\binom{|N_i|}{r}$ eligible sets and then using $|N_i| \leq d$ proves both inequalities. If $d = 0$, every neighborhood is empty and all displayed quantities are zero, so the same conclusion holds. $\square$

Justification. This identity charges overlap once through out-degree; in-neighborhood combinatorics are already contained in A_d . *Gap.* The stated SNIPE variance analysis in [1] bounds neighborhood size and overlap separately; this identity shows how to charge out-degree only once. *Consumer.* It is the load-bearing covariance count for the degree-sharp SNIPE variance bound on both bounded classes.

Theorem 1 (thm:degree-frontier). Fix $\beta \geq 1$, $p \in (0, 1)$, $B \geq 0$, integers $n \geq 1$, and $0 \leq d \leq n$. There are $0 < c_{\beta,p} \leq C_{\beta,p} < \infty$ such that

$$c_{\beta,p} B^2 \min \left\{ 1, \frac{d \binom{d}{k_*}}{n} \right\} \leq R_n^*(d, \beta, p, B) \leq \sup_{(G_n, c) \in \mathcal{M}_{n,d,\beta}(B)} \mathbb{E}_Z[(\hat{\tau}_n^{\text{up}} - \tau_n)^2] \leq C_{\beta,p} B^2 \min \left\{ 1, \frac{d \binom{d}{k_*}}{n} \right\}.$$

In the nonsaturated regime $d \binom{d}{k_*} / n \lesssim_{\beta,p} 1$, unclipped SNIPE attains the same rate. The upper bound uses $A_i = \sum_{r=1}^{\min\{\beta, |N_i|\}} \binom{|N_i|}{r} \Delta_r(p)^2 / [p(1-p)]^r \leq A_d$ and the single-out-degree overlap count. For $B > 0$ and $d \geq 1$, the block mixtures have TTE separation $2\rho\delta$ and squared Hellinger distance at most $4\pi^2 m \delta^2 / (B^2 A_d)$; since $1/2 \leq \rho \leq 1$ and $A_d/m = (dA_d/n)/\rho$, this yields the stated lower branch without a divisibility condition.

Proof. By def : two – class – minimax – risks, $R_{n,\ell_1}^*(d, \beta, p, B) = R_n^*(d, \beta, p, B)$. Apply Theorem thm : bounded – outcome – degree – frontier to the coefficient-mass class. It gives constants depending only on β, p for which

$$c_{\beta,p} B^2 \min \left\{ 1, \frac{dA_d}{n} \right\} \leq R_n^*(d, \beta, p, B) \leq \sup_{(G_n, c) \in \mathcal{M}_{n,d,\beta}(B)} \mathbb{E}_Z[(\hat{\tau}_n^{\text{up}} - \tau_n)^2] \leq C_{\beta,p} B^2 \min \left\{ 1, \frac{dA_d}{n} \right\}.$$

The same theorem states, uniformly for $0 \leq d \leq n$, that

$$dA_d \asymp_{\beta,p} d \binom{d}{k_*}.$$

Absorbing the two comparison constants into $c_{\beta,p}$ and $C_{\beta,p}$ proves the displayed frontier in the present statement. The theorem also proves design unbiasedness of unclipped SNIPE and the bound

$$\sup_{(G_n, c) \in \mathcal{M}_{n,d,\beta}(B)} \mathbb{E}_Z[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2] \leq B^2 \frac{dA_d}{n}.$$

Consequently, whenever $d \binom{d}{k_*} / n \lesssim_{\beta,p} 1$, this risk is within a constant depending only on β, p of the minimax lower bound, so unclipped SNIPE attains the nonsaturated rate.

For completeness, all additional assertions in the statement are direct specializations of the stronger theorem. Its upper-bound calculation defines

$$A_i = \sum_{r=1}^{\min\{\beta, |N_i|\}} \binom{|N_i|}{r} \frac{\Delta_r(p)^2}{[p(1-p)]^r} \leq A_d$$

and invokes the single-out-degree overlap count. Its least-favourable block mixtures have TTEs $\rho\delta$ and $-\rho\delta$, hence separation $2\rho\delta$, and their squared Hellinger distance is at most $4\pi^2 m \delta^2 / (B^2 A_d)$. Finally, it establishes

$$\frac{1}{2} \leq \rho \leq 1, \quad \frac{A_d}{m} = \frac{dA_d/n}{\rho},$$

which yields the lower branch without requiring $d \mid n$. This proves every claim. $\square$

Justification. This coefficient-mass specialization records the original public frontier in its prior notation and is now discharged by the stronger simultaneous two-class theorem. *Gap.* The stated upper bound in [1] is coarser by one overlap factor and does not expose the cancellation-sensitive energy A_d ; its Gaussian-prior lower bound is not a bounded-class converse. *Consumer.* Readers retaining the published $Y_{\max}$ coefficient-mass normalization can use this theorem directly, while the stronger headline shows the same rate persists under mere uniform outcome boundedness.

Theorem 2 (thm:sharp-local-linear-constant-and-representers). Fix $\beta \geq 1$, $p \in (0, 1)$, and $B > 0$. For every index t , let n_t and d_t be positive integers such that $d_t \mid n_t$, write $m_t := n_t/d_t$, and let G_t be the fixed directed graph that is the disjoint union of m_t complete directed blocks V_b of size d_t , including all self-loops. Then, for every t ,

$$R_t^{\text{loc,lin}} = \frac{B^2 A_{d_t}}{m_t} = \frac{B^2 d_t A_{d_t}}{n_t}.$$

In particular, $R_t^{\text{loc,lin}}/(B^2 d_t A_{d_t}/n_t) = 1$ for every t . More precisely, put $Z_S := \prod_{j \in S} Z_j$ and $Z_b := (Z_j)_{j \in V_b}$, and for every block V_b define

$$\Psi_{b,t}(w) := \max_{\substack{(\varepsilon_i, S_i)_{i \in V_b} \\ \varepsilon_i \in \{-1, 1\}, S_i \subseteq V_b, |S_i| \leq \bar{\beta}_{d_t}}} \mathbb{E}_Z \left[\left(\sum_{i \in V_b} \varepsilon_i \{w_{i,t}(Z_b) Z_{S_i} - \mathbf{1}(S_i \neq \emptyset)\} \right)^2 \right].$$

Then, for every t ,

$$\sup_{c \in \mathcal{M}_t(G_t)} \mathbb{E}_Z [(\hat{\tau}_{w,t} - \tau_{n_t})^2] = \frac{B^2}{n_t^2} \sum_{b=1}^{m_t} \Psi_{b,t}(w), \quad \Psi_{b,t}(w) \geq d_t^2 A_{d_t}.$$

A sequence in $\mathcal{E}_t^{\text{loc,lin}}$ is asymptotically minimax if and only if

$$\frac{1}{m_t d_t^2 A_{d_t}} \sum_{b=1}^{m_t} \{\Psi_{b,t}(w) - d_t^2 A_{d_t}\} \rightarrow 0.$$

The condition

$$\frac{1}{n_t A_{d_t}} \sum_{i \in V_{n_t}} \mathbb{E}_Z [(w_{i,t}(Z_{b(i)}) - g_{d_t}(Z_{b(i)}))^2] \rightarrow 0$$

is sufficient for asymptotic minimaxity. Suppose in addition that $d_t \rightarrow \infty$. This sufficient condition is not necessary, even up to relabeling coordinates within blocks. There are asymptotically minimax weights for which the last display equals 2 for every sufficiently large t .

Proof. Fix an index t , and abbreviate $n = n_t$, $d = d_t$, $m = m_t = n/d$, $A = A_d$, $g = g_d$, and $\mathcal{P} = \mathcal{P}_d$. For $S \subseteq V_b$, write $Z_S = \prod_{j \in S} Z_j$, and identify the coordinates of each block with $\{1, \dots, d\}$. By def : local – linear – class, every admissible weight $w_i := w_{i,t}$ satisfies

$$\mathbb{E}_Z [w_i(Z_b) f(Z_b)] = L_d(f), \quad f \in \mathcal{P}.$$

Indeed, this follows by linearity from the constant and nonempty-monomial moment restrictions in that definition. By lem : block – energy – representer, the same identity holds with g in place of w_i , and $\|g\|_2^2 = A$. Consequently every admissible weight has the orthogonal decomposition

$$w_i = g + u_i, \quad u_i \perp \mathcal{P}. \quad (1)$$

All norms below are in the block space $L^2(P_Z)$.

We first construct an attaining estimator. Set $w_i = g$ for every unit. For $S \subseteq V_b$ with $|S| \leq \bar{\beta}_d$, define

$$X_S := g(Z_b) Z_S - \mathbf{1}\{S \neq \emptyset\}.$$

If $S = \emptyset$, then $\|X_S\|_2^2 = A$. If $S \neq \emptyset$, the representer identity, $Z_S^2 = Z_S$, and $0 \leq Z_S \leq 1$ give

$$\begin{aligned} \|X_S\|_2^2 &= \mathbb{E}_Z [g(Z_b)^2 Z_S] - 2\mathbb{E}_Z [g(Z_b) Z_S] + 1 \\ &= \mathbb{E}_Z [g(Z_b)^2 Z_S] - 1 \leq \mathbb{E}_Z [g(Z_b)^2] = A. \end{aligned} \quad (2)$$

For an admissible unit-level polynomial $f_i = \sum_S c_{i,S} Z_S$, the triangle inequality and the coefficient-mass restriction in def : local – linear – class therefore imply

$$\|gf_i - L_d(f_i)\|_2 \leq \sum_S |c_{i,S}| \|X_S\|_2 \leq B\sqrt{A}. \quad (3)$$

Under ass : bernoulli – design, assignment subvectors belonging to distinct blocks are independent. Each block error is centered by the defining moment restrictions. Hence block variances add, and applying the triangle inequality within each block to (3) yields

$$\sup_{c \in \mathcal{M}_t(G_t)} \mathbb{E}_Z [(\hat{\tau}_g - \tau_n)^2] \leq \frac{1}{n^2} \sum_{b=1}^m (dB\sqrt{A})^2 = \frac{B^2 A}{m}. \quad (4)$$

For the matching converse, take the admissible constant schedule $f_i \equiv B$ for every unit. It has coefficient mass B and $L_d(f_i) = 0$. For arbitrary admissible weights, (1), together with $g \in \mathcal{P}$ and $u_i \perp \mathcal{P}$, gives in each block

$$\left\| \sum_{i \in V_b} w_i \right\|_2^2 = \left\| dg + \sum_{i \in V_b} u_i \right\|_2^2 = d^2 A + \left\| \sum_{i \in V_b} u_i \right\|_2^2 \geq d^2 A. \quad (5)$$

Independence across blocks then implies

$$\sup_{c \in \mathcal{M}_t(G_t)} \mathbb{E}_Z[(\hat{\tau}_w - \tau_n)^2] \geq \frac{B^2}{n^2} \sum_{b=1}^m d^2 A = \frac{B^2 A}{m}. \quad (6)$$

Combining (4) and (6), and restoring subscripts, proves for every t that

$$R_t^{\text{loc,lin}} = \frac{B^2 A_{d_t}}{m_t} = \frac{B^2 d_t A_{d_t}}{n_t}. \quad (7)$$

Because $B > 0$, $d_t \geq 1$, and $A_{d_t} > 0$ by $\Delta_1(p) = 1$, the denominator in the asserted ratio is positive; thus (7) also proves that this ratio equals one.

We next prove the exact finite extremal formula. For a fixed block V_b , define its unscaled centered error by

$$E_b(c, w) := \sum_{i \in V_b} \{w_i(Z_b) f_i(Z_b) - L_d(f_i)\}.$$

As a function of the block coefficient vectors $(c_{i,S})_{i \in V_b}$, the map $\|E_b(c, w)\|_2^2$ is convex. Each unit's feasible coefficient vector lies in a finite-dimensional ℓ_1 -ball of radius B , which is the convex hull of the vectors $\varepsilon_i B e_{S_i}$, where $\varepsilon_i \in \{-1, 1\}$ and $|S_i| \leq \bar{\beta}_d$. The product of the unit-level balls is the convex hull of the corresponding product points. Jensen's inequality therefore bounds the value at every feasible coefficient array by the largest value at such a product point. At that point,

$$E_b(c, w) = B \sum_{i \in V_b} \varepsilon_i \{w_i(Z_b) Z_{S_i} - \mathbf{1}(S_i \neq \emptyset)\}. \quad (8)$$

Conversely, every such product point is feasible, so this bound is attained. The block errors are centered and independent under $\text{ass : bernoulli} - \text{design}$, and the coefficient vectors may be chosen independently across blocks. Thus the supremum of the sum of block variances is the sum of their suprema. Equation (8) gives exactly

$$\sup_{c \in \mathcal{M}_t(G_t)} \mathbb{E}_Z[(\hat{\tau}_{w,t} - \tau_{n_t})^2] = \frac{B^2}{n_t^2} \sum_{b=1}^{m_t} \Psi_{b,t}(w). \quad (9)$$

In the maximization defining $\Psi_{b,t}$, choose $S_i = \emptyset$ and $\varepsilon_i = 1$ for every $i \in V_b$. Equation (5) then yields

$$\Psi_{b,t}(w) \geq d_t^2 A_{d_t}. \quad (10)$$

This proves all finite assertions.

By definition, a sequence is asymptotically minimax precisely when its worst-case risk divided by $R_t^{\text{loc,lin}}$ tends to one. Equations (7), (9), and (10) imply, without requiring $d_t \rightarrow \infty$,

$$\frac{\sup_c \mathbb{E}_Z[(\hat{\tau}_{w,t} - \tau_{n_t})^2]}{R_t^{\text{loc,lin}}} = 1 + \frac{1}{m_t d_t^2 A_{d_t}} \sum_{b=1}^{m_t} \{\Psi_{b,t}(w) - d_t^2 A_{d_t}\}. \quad (11)$$

Every summand in the last numerator is nonnegative by (10). Hence the risk ratio tends to one if and only if the normalized excess in the theorem tends to zero. This proves the asserted characterization for every admissible sequence.

We now prove the sufficient condition, still without assuming that d_t diverges. Suppress t again and set

$$a_b^2 := \sum_{i \in V_b} \|u_i\|_2^2, \quad U := \sum_{b=1}^m a_b^2.$$

Every admissible polynomial satisfies $|f_i(Z_b)| \leq B$, because its raw coefficient mass is at most B . Therefore Cauchy–Schwarz over the units gives

$$\left\| \sum_{i \in V_b} u_i f_i \right\|_2 \leq B \sum_{i \in V_b} \|u_i\|_2 \leq B\sqrt{d} a_b. \quad (12)$$

By (3), the part of the block error involving g has norm at most $dB\sqrt{A}$. Combining this with (12), summing squared block bounds, and using $\sum_b a_b \leq \sqrt{mU}$, we obtain

$$1 \leq \frac{\sup_c \mathbb{E}_Z[(\hat{\tau}_w - \tau_n)^2]}{B^2 A/m} \leq 1 + 2\sqrt{\frac{U}{mdA}} + \frac{U}{mdA} = \left(1 + \sqrt{\frac{U}{nA}}\right)^2. \quad (13)$$

The lower inequality is (6). By (1),

$$U = \sum_{i \in V_n} \mathbb{E}_Z[(w_i(Z_{b(i)}) - g(Z_{b(i)}))^2].$$

Thus the normalized L^2 condition in the theorem makes the right-hand side of (13) tend to one. The lower bound in (13) then proves asymptotic minimaxity.

It remains to prove non-necessity under the additional hypothesis $d_t \rightarrow \infty$. Since β is fixed, for every sufficiently large t ,

$$\dim(\mathcal{P}_{d_t}) = \sum_{r=0}^{\beta} \binom{d_t}{r} < 2^{d_t} = \dim L^2(P_Z). \quad (14)$$

The last equality holds because $p \in (0, 1)$ gives full support on the block assignment space. Hence $\mathcal{P}_{d_t}^\perp$ contains a nonzero function. Also $A_{d_t} > 0$ because $\Delta_1(p) = 1$. Choose $v_{d_t} \in \mathcal{P}_{d_t}^\perp$ normalized so that

$$\|v_{d_t}\|_2^2 = d_t A_{d_t}. \quad (15)$$

For every block and all sufficiently large t , set

$$u_1 = v_{d_t}, \quad u_2 = -v_{d_t}, \quad u_i = 0 \quad (i \geq 3), \quad w_i = g_{d_t} + u_i. \quad (16)$$

There are at least two units per block eventually because $d_t \rightarrow \infty$, and the orthogonality in (15) verifies every moment restriction in def : local – linear – class. For arbitrary admissible unit polynomials, the additional block error contributed by (16) is $v_{d_t}(f_1 - f_2)$, whose norm is at most $2B\sqrt{d_t A_{d_t}}$. The canonical g_{d_t} -part has norm at most $d_t B\sqrt{A_{d_t}}$ by (3). Consequently

$$1 \leq \frac{\sup_c \mathbb{E}_Z[(\hat{\tau}_w - \tau_{n_t})^2]}{B^2 A_{d_t}/m_t} \leq \left(1 + \frac{2}{\sqrt{d_t}}\right)^2 \rightarrow 1. \quad (17)$$

Thus these weights are asymptotically minimax. On the other hand, (15) and (16) give, for every sufficiently large t ,

$$\frac{1}{n_t A_{d_t}} \sum_{i \in V_{n_t}} \|w_i - g_{d_t}\|_2^2 = \frac{m_t(2d_t A_{d_t})}{m_t d_t A_{d_t}} = 2. \quad (18)$$

A relabeling of coordinates within a block preserves the product Bernoulli law, the symmetric function g_{d_t} , and the sum of squared distances in (18). It therefore cannot make the sufficient condition hold. This proves the non-necessity and the distance-exactly-two witness under precisely the stated divergence hypothesis. $\square$

Justification. This theorem resolves the former structural question: on fixed complete directed blocks in the ℓ_1 class it gives the exact finite minimax risk, the exact per-block extremal functional, and the complete asymptotic optimality characterization. *Gap.* The general Riesz framework of [2] does not determine this finite-block minimax constant or all asymptotically minimax local-weight collections. *Consumer.* Readers can use $B^2 A_{d_t}/m_t$ as the exact block-local linear benchmark, $\Psi_{b,t}$ as a certificate for proposed weights, and the distance-two witness as a warning that normalized average L^2 -closeness to SNIPE is sufficient but not necessary when $d_t \rightarrow \infty$.

Theorem 3 (thm:bounded-outcome-degree-frontier). *Fix $\beta \geq 1$, $p \in (0, 1)$, $B \geq 0$, integers $n \geq 1$, and $0 \leq d \leq n$. The coefficient-mass model is contained in the uniformly bounded-outcome model, and the inclusion is strict whenever $B > 0$ and $d \geq 1$. There are $0 < c_{\beta,p} \leq C_{\beta,p} < \infty$ such that*

$$c_{\beta,p} B^2 \min\left\{1, \frac{dA_d}{n}\right\} \leq R_{n,\ell_1}^*(d, \beta, p, B) = R_n^*(d, \beta, p, B) \leq R_{n,\infty}^*(d, \beta, p, B) \leq \sup_{(G_n, c) \in \mathcal{M}_{n,d,\beta}^\infty(B)} \mathbb{E}_Z[(\hat{\tau}_{n,\infty}^{\text{up}} - \tau_n)^2] \leq C_{\beta,p} B^2 \min\left\{1, \frac{dA_d}{n}\right\}.$$

Moreover,

$$\sup_{(G_n, c) \in \mathcal{M}_{n,d,\beta}(B)} \mathbb{E}_Z[(\hat{\tau}_n^{\text{up}} - \tau_n)^2] \leq C_{\beta,p} B^2 \min\left\{1, \frac{dA_d}{n}\right\},$$

and on either model class unclipped SNIPE is design-unbiased and has worst-case mean squared error at most $B^2 dA_d/n$. For every $0 \leq d \leq n$, $dA_d \asymp_{\beta,p} d(k_*)^d$, so both minimax risks are also $\asymp_{\beta,p} B^2 \min\{1, d(k_*)^d/n\}$, and unclipped SNIPE attains the nonsaturated branch. If $1 \leq d \leq n$ and $d \mid n$, put $m = n/d$ and take the disjoint union of m complete directed d -blocks, including loops. Then canonical unclipped SNIPE has the exact global worst-case risk on both classes,

$$\sup_{(G_n, c) \in \mathcal{M}_{n,d,\beta}(B)} \mathbb{E}_Z[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2] = \sup_{(G_n, c) \in \mathcal{M}_{n,d,\beta}^\infty(B)} \mathbb{E}_Z[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2] = \frac{B^2 A_d}{m} = \frac{B^2 dA_d}{n},$$

so the constant one in the global unclipped variance bound is sharp. The upper-bound calculation uses

$$A_i = \sum_{r=1}^{\min\{\beta, |N_i|\}} \binom{|N_i|}{r} \frac{\Delta_r(p)^2}{[p(1-p)]^r} \leq A_d$$

and charges out-degree only once. For $B > 0$ and $d \geq 1$, the block mixtures in `def : block – family` lie in the smaller class, have TTE separation $2\rho\delta$, and have squared Hellinger distance at most $4\pi^2 m\delta^2/(B^2 A_d)$; furthermore $1/2 \leq \rho \leq 1$ and $A_d/m = (dA_d/n)/\rho$.

Proof. We first dispose of the boundary cases. If $B = 0$, all potential outcomes in either class vanish, so the observed data, the TTE, both clipped estimators, and both minimax risks are zero. If $d = 0$, `ass : bounded – degree` makes every N_i empty. Every potential outcome is then assignment-invariant, $\tau_n = 0$, and `def : zero – degree – conventions` gives $A_0 = 0$ and zero SNIPE score. All conclusions follow. Henceforth assume $B > 0$ and $1 \leq d \leq n$.

Comparison of the model classes. If $c \in \mathcal{C}_{n,d,\beta}(B)$, then for every $i \in V_n$ and every assignment z ,

$$|Y_i(z)| \leq \sum_{S \subseteq N_i} |c_{i,S}| \prod_{j \in S} z_j \leq \sum_{S \subseteq N_i} |c_{i,S}| \leq B.$$

The low-order restriction is common to the two classes. Therefore

$$\mathcal{M}_{n,d,\beta}(B) \subseteq \mathcal{M}_{n,d,\beta}^\infty(B), \quad R_{n,\ell_1}^*(d, \beta, p, B) \leq R_{n,\infty}^*(d, \beta, p, B).$$

The inclusion is strict. Indeed, take a graph having the single self-loop at unit 1 and no other edge, set

$$Y_1(z) = B(2z_1 - 1), \quad Y_i(z) = 0 \quad (i \neq 1).$$

This schedule has order one and satisfies $|Y_i(z)| \leq B$. Its two nonzero raw coefficients are $c_{1,\emptyset} = -B$ and $c_{1,\{1\}} = 2B$, whose absolute mass is $3B > B$. Thus it belongs to the bounded-outcome class but not the coefficient-mass class.

Unbiasedness and the common variance bound. Define

$$W_i := Y_i^{\text{obs}} g_{i,\beta,p}(Z), \quad q_i := |N_i|.$$

If $q_i = 0$, both the score and unit i 's causal contrast vanish. If $q_i \geq 1$, relabel the coordinates in N_i . The representer identity in `lem : block – energy – representer`, applied in dimension q_i , and `ass : low – order imply`

$$\mathbb{E}_Z[W_i] = Y_i(\mathbf{1}) - Y_i(\mathbf{0}).$$

Summing over i proves

$$\mathbb{E}_Z[\widehat{\tau}_n^{\text{SNIPE}}] = \tau_n$$

on both classes. Centered-monomial orthogonality under $\text{ass} : \text{bernoulli} - \text{design}$ also gives

$$\mathbb{E}_Z[g_{i,\beta,p}(Z)^2] = A_i := \sum_{r=1}^{\min\{\beta, q_i\}} \binom{q_i}{r} \frac{\Delta_r(p)^2}{[p(1-p)]^r} \leq A_d,$$

because $q_i \leq d$ and $\binom{q}{r}$ is nondecreasing in q for fixed r .

Every centered function of the binary vector Z_{N_i} has a unique Bernoulli Fourier expansion

$$W_i - \mathbb{E}_Z W_i = \sum_{\emptyset \neq S \subseteq N_i} a_{i,S} \prod_{j \in S} (Z_j - p).$$

On the larger class, $|Y_i(Z)| \leq B$. Orthogonality consequently yields

$$\sum_{\emptyset \neq S \subseteq N_i} a_{i,S}^2 [p(1-p)]^{|S|} = \text{Var}_Z(W_i) \leq \mathbb{E}_Z[W_i^2] \leq B^2 A_i \leq B^2 A_d.$$

For a nonempty set S , let $I_S := \{i \in V_n : S \subseteq N_i\}$. Choose any $j \in S$. The out-degree restriction in $\text{ass} : \text{bounded} - \text{degree}$, equivalently the one-coordinate charge underlying $\text{lem} : \text{overlap} - \text{count}$, gives

$$|I_S| \leq \#\{i \in V_n : j \in N_i\} \leq d.$$

Another use of orthogonality, followed by Cauchy-Schwarz for the sum over I_S , gives

$$\begin{aligned} \text{Var}_Z \left(\sum_{i \in V_n} W_i \right) &= \sum_{S \neq \emptyset} [p(1-p)]^{|S|} \left(\sum_{i \in I_S} a_{i,S} \right)^2 \\ &\leq d \sum_{i \in V_n} \sum_{\emptyset \neq S \subseteq N_i} a_{i,S}^2 [p(1-p)]^{|S|} \\ &\leq ndB^2 A_d. \end{aligned}$$

Design unbiasedness therefore implies, on either class,

$$\mathbb{E}_Z[(\widehat{\tau}_n^{\text{SNIPE}} - \tau_n)^2] \leq B^2 \frac{dA_d}{n}.$$

On the bounded-outcome class,

$$|\tau_n| \leq \frac{1}{n} \sum_{i \in V_n} \{|Y_i(\mathbf{1})| + |Y_i(\mathbf{0})|\} \leq 2B.$$

Projection onto $[-2B, 2B]$ cannot increase distance to τ_n , and the squared distance between two points in this interval is at most $16B^2$. Hence

$$\sup_{(G_n, c) \in \mathcal{M}_{n,d,\beta}^\infty(B)} \mathbb{E}_Z[(\widehat{\tau}_{n,\infty}^{\text{up}} - \tau_n)^2] \leq B^2 \min \left\{ \frac{dA_d}{n}, 16 \right\} \leq 16B^2 \min \left\{ 1, \frac{dA_d}{n} \right\}.$$

On the coefficient-mass class, each unit contrast has absolute value at most B , so $|\tau_n| \leq B$. Projection onto $[-B, B]$ similarly gives

$$\sup_{(G_n, c) \in \mathcal{M}_{n,d,\beta}(B)} \mathbb{E}_Z[(\hat{\tau}_n^{\text{up}} - \tau_n)^2] \leq 4B^2 \min\left\{1, \frac{dA_d}{n}\right\}.$$

Exact complete-block worst-case risk. Suppose $d \mid n$, put $m = n/d$, and take the disjoint union of m complete directed blocks of size d , including loops. The variance inequality just proved gives the upper bound

$$B^2 \frac{dA_d}{n} = \frac{B^2 A_d}{m}$$

for the supremum on either class. For the reverse inequality, take the fixed schedule $Y_i(z) \equiv B$ for every unit. It belongs to both classes and has $\tau_n = 0$. All units in block b share the score $g_d(Z_b)$, whence

$$\hat{\tau}_n^{\text{SNIPE}} = \frac{Bd}{n} \sum_{b=1}^m g_d(Z_b).$$

The block assignments are independent, $\mathbb{E}_Z[g_d(Z_b)] = 0$, and $\text{lem : block - energy - representer}$ gives $\mathbb{E}_Z[g_d(Z_b)^2] = A_d$. Therefore

$$\mathbb{E}_Z[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2] = \frac{B^2 d^2 m A_d}{n^2} = \frac{B^2 A_d}{m} = \frac{B^2 d A_d}{n}.$$

Thus both global suprema equal the claimed value, proving sharpness of the constant one.

Least-favourable family and converse. Use $\text{def : block - family}$ and abbreviate

$$\kappa := \min\left\{(2H_{\beta,p})^{-1}, (4\pi)^{-1}\right\}, \quad \delta = \kappa B \min\left\{1, \sqrt{A_d/m}\right\}.$$

The complete active blocks have in-degree and out-degree exactly d , including the loop, and inactive vertices have both degrees zero. Thus the graph is in $\mathcal{G}_{n,d}$. By $\text{def : exposed - order}$, all coefficients of g_d , hence of $h_d = g_d/A_d$, above order k_* vanish. The schedules therefore have order at most $k_* \leq \beta$. For active $i \in V_b$, $\text{lem : block - energy - representer}$, $|U_b| \leq B/2$, and $\kappa H_{\beta,p} \leq 1/2$ imply

$$\sum_{T \subseteq V_b} |c_{i,T}| \leq |U_b| + \delta \sum_T |h_{d,T}| \leq \frac{B}{2} + \kappa B H_{\beta,p} \leq B.$$

Inactive outcomes vanish. Every schedule in either prior's support consequently belongs to $\mathcal{M}_{n,d,\beta}(B)$, and hence also to $\mathcal{M}_{n,d,\beta}^\infty(B)$.

For active $i \in V_b$,

$$Y_i(z) = U_b + \sigma \delta h_d(z_{V_b}).$$

Since $h_d(\mathbf{1}_d) - h_d(\mathbf{0}_d) = 1$, every schedule under Π_σ has

$$\tau_n = \sigma \rho \delta.$$

The two prior mixtures thus have TTE separation $2\rho\delta$.

It remains to bound their statistical distance. Let $\psi_s := \sqrt{f_s}$, extended by zero outside $[-s, s]$. It is weakly differentiable, and direct calculation gives

$$\|\psi'_s\|_2^2 = \frac{\pi^2}{4s^3} \int_{-s}^s \sin^2\left(\frac{\pi u}{2s}\right) du = \frac{\pi^2}{4s^2}.$$

For any real a, b , the fundamental theorem of calculus followed by Minkowski's integral inequality gives

$$\|\psi_s(\cdot - a) - \psi_s(\cdot - b)\|_2 \leq |a - b| \|\psi'_s\|_2.$$

Conditional on a block assignment $Z_b = z$, the repeated observed block outcome has a translate of density f_s , and the translations under the two signs differ by $2\delta h_d(z)$. Under the convention

$$H^2(P, Q) := \int (\sqrt{dP} - \sqrt{dQ})^2,$$

the one-block squared Hellinger distance is therefore bounded by

$$\mathbb{E}_Z \left[4\delta^2 h_d(Z)^2 \|\psi'_s\|_2^2 \right] = \frac{\pi^2 \delta^2}{s^2} \mathbb{E}_Z [h_d(Z)^2] = \frac{4\pi^2 \delta^2}{B^2 A_d},$$

where $s = B/2$ and $\text{lem} : \text{block} - \text{energy} - \text{representer}$ supplies $\mathbb{E}_Z[h_d(Z)^2] = A_d^{-1}$. Hellinger affinity multiplies over the m independent blocks. Since $1 - \prod_{b=1}^m a_b \leq \sum_{b=1}^m (1 - a_b)$ for $a_b \in [0, 1]$,

$$H^2(P_+, P_-) \leq \frac{4\pi^2 m \delta^2}{B^2 A_d} \leq 4\pi^2 \kappa^2 \leq \frac{1}{4}.$$

Cauchy–Schwarz applied to the densities gives $\|P_+ - P_-\|_{\text{TV}} \leq H(P_+, P_-)$, and hence

$$\|P_+ - P_-\|_{\text{TV}} \leq \frac{1}{2}.$$

For any estimator, thresholding it at zero defines a test of the two signs. Put $\theta := \rho\delta$. On a testing error, the squared estimation error is at least θ^2 . The sum of the two testing-error probabilities is at least $1 - \|P_+ - P_-\|_{\text{TV}}$. Consequently

$$\max_{\sigma \in \{-1, 1\}} \mathbb{E}_{P_\sigma} [(\hat{\tau}_n - \sigma\theta)^2] \geq \frac{\theta^2}{2} \{1 - \|P_+ - P_-\|_{\text{TV}}\} \geq \frac{\rho^2 \delta^2}{4}.$$

Each prior is supported on deterministic schedules in the smaller model class, so its integrated risk cannot exceed the corresponding worst-case risk. This inequality therefore lower-bounds both minimax risks.

Finally, $m = \lfloor n/d \rfloor$ and $1 \leq d \leq n$ imply

$$\frac{1}{2} \leq \rho = \frac{md}{n} \leq 1, \quad \frac{A_d}{m} = \frac{dA_d/n}{\rho}.$$

Using the definition of δ ,

$$R_{n, \ell_1}^*(d, \beta, p, B) \geq \frac{\kappa^2}{16} B^2 \min \left\{ 1, \frac{dA_d}{n} \right\}.$$

Class inclusion gives the same lower bound for $R_{n, \infty}^*$. Together with the upper bounds, this proves the A_d -scale sandwich with $c_{\beta, p} = \kappa^2/16$ and, for example, $C_{\beta, p} = 16$.

For $d \geq 1$, $\text{lem} : \text{block} - \text{energy} - \text{representer}$ gives $A_d \asymp_{\beta, p} \binom{d}{k_*}$. For $d = 0$, $\text{def} : \text{zero} - \text{degree} - \text{conventions}$ gives $dA_d = 0 = d \binom{d}{k_*}$. Hence, uniformly for $0 \leq d \leq n$,

$$dA_d \asymp_{\beta, p} d \binom{d}{k_*}.$$

Substitution proves the binomial frontier. The unclipped risk bound $B^2 dA_d/n$ is of the same order as the minimax risk whenever $d \binom{d}{k_*}/n \lesssim_{\beta, p} 1$, completing the proof. $\square$

Justification. This theorem gives the simultaneous matched design-based minimax frontier over the ℓ_1 coefficient-mass class and the larger uniformly bounded-outcome class, with the exact energy A_d leading, one clipped attaining estimator, a noiseless least-favourable family, and the exact worst-case risk of canonical unclipped SNIPE on complete blocks. *Gap.* Theorem 1 of [1] has a coarser stated worst-case variance bound, while its Gaussian-prior Theorem 2 is not a converse for either bounded class. This theorem supplies the missing cancellation-sensitive two-class frontier without claiming to improve that unrestricted Gaussian-prior result. *Consumer.* For the singleton-cluster candidate $N = 1$, $C = d_{\max}$ in the clustering-selection procedure of [4], the theorem replaces the coarse unit-Bernoulli calibration by dA_d/n ; at $p = 1/2$, $\beta = 1$, the linear-regime threshold is $n \gg d^2$. This recalibrates that candidate, not the paper's general clustered-design bound.

Theorem 4 (thm:fair-coin-energy-frontier). *Fix $n \geq 1$, $1 \leq d \leq n$, $\beta \geq 1$, $B \geq 0$, and $p = 1/2$. Then*

$$\Delta_r(1/2) = \begin{cases} 2^{1-r}, & r \text{ odd}, \\ 0, & r \text{ even}, \end{cases} \quad A_d = 4 \sum_{\substack{1 \leq r \leq \min\{\beta, d\} \\ r \text{ odd}}} \binom{d}{r}.$$

Consequently $k_\star$ is the largest odd integer no greater than $\min\{\beta, d\}$. For $\beta = 1$, $A_d = 4d$ and

$$R_n^\star(d, 1, 1/2, B) = R_{n, \ell_1}^\star(d, 1, 1/2, B) \asymp R_{n, \infty}^\star(d, 1, 1/2, B) \asymp B^2 \min\left\{1, \frac{d^2}{n}\right\}.$$

If additionally $d \mid n$, canonical unclipped SNIPE has exact worst-case mean squared error $4B^2d^2/n$ on both model classes.

Proof. At $p = 1/2$, for every integer $r \geq 1$,

$$\Delta_r(1/2) = 2^{-r} \{1 - (-1)^r\} = \begin{cases} 2^{1-r}, & r \text{ odd}, \\ 0, & r \text{ even}. \end{cases}$$

Substitute this identity and $p(1-p) = 1/4$ into $\text{def} : \text{block} - \text{score} - \text{energy}$. For odd r ,

$$\frac{\Delta_r(1/2)^2}{[p(1-p)]^r} = \frac{2^{2-2r}}{4^{-r}} = 4,$$

whereas for even r the summand is zero. Therefore

$$A_d = 4 \sum_{\substack{1 \leq r \leq \min\{\beta, d\} \\ r \text{ odd}}} \binom{d}{r}.$$

Because $d \geq 1$ and $\beta \geq 1$, the exposed set is nonempty, and $\text{def} : \text{exposed} - \text{order}$ now shows that $k_\star$ is the largest odd integer not exceeding $\min\{\beta, d\}$.

If $\beta = 1$, only $r = 1$ occurs, so

$$A_d = 4 \binom{d}{1} = 4d.$$

The identities in $\text{def} : \text{two} - \text{class} - \text{minimax} - \text{risks}$ and the two-class frontier in Theorem thm : bounded - outcome - degree - frontier then give

$$R_n^\star(d, 1, 1/2, B) = R_{n, \ell_1}^\star(d, 1, 1/2, B) \asymp R_{n, \infty}^\star(d, 1, 1/2, B) \asymp B^2 \min\left\{1, \frac{d^2}{n}\right\}.$$

Finally, if $d \mid n$, the exact complete-block assertion of that theorem gives

$$\sup_{(G_n, c) \in \mathcal{M}_{n, d, 1}(B)} \mathbb{E}_Z[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2] = \sup_{(G_n, c) \in \mathcal{M}_{n, d, 1}^\infty(B)} \mathbb{E}_Z[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2] = \frac{B^2 d A_d}{n} = \frac{4B^2 d^2}{n}.$$

This is the claimed exact worst-case mean squared error. $\square$

Justification. This public fair-coin theorem makes the exposed-order cancellation and exact energy transparent: only odd orders contribute, and $\beta = 1$ gives the motivating d^2/n frontier. *Gap.* The general bounds in [1] do not state this cancellation-sensitive minimax specialization or the exact fair-coin energy identity. *Consumer.* It gives the most legible known-graph additive-interference benchmark adjacent to [3] and the concrete singleton-cluster calibration used in comparison with [4].

7 Interpretation

The exact local energy A_d measures the squared norm of the Bernoulli representer for the all-one-versus-all-zero contrast, while the extra factor d records how many outcome neighborhoods one assignment coordinate can affect. At $p = 1/2$, even-order contrast terms vanish from this representer and A_d is four times the sum of the eligible odd-order binomial counts; for $\beta = 1$, the matched linear branch is therefore d^2/n . Complete blocks show that the unit upper constant one for canonical unclipped SNIPE is genuinely attained, not merely an artifact of the proof. Within the narrower block-local linear problem, g_d attains the exact minimax constant, but asymptotic minimaxity need not force every unit weight to approach g_d : opposite perturbations in $\mathcal{P}_d^\perp$ can cancel within blocks while the normalized aggregate L^2 distance remains two.

Technical internal limitation (diagnostic only)

The main upper bound over the uniformly bounded-outcome class uses only $|Y_i(z)| \leq B$; this yields $|\tau_n| \leq 2B$, so the attaining estimator is projected onto $[-2B, 2B]$. The converse is inherited from the smaller ℓ_1 class because the continuous-baseline least-favourable schedules have coefficient mass at most B . The exact block-local linear theorem is not enlarged to bounded outcomes: its within-block triangle argument and finite ℓ_1 -extreme-point formula for $\Psi_{b,t}$ genuinely use coefficient-mass boundedness. The lower bound uses Hellinger distance because compactly supported translates of f_s may have infinite one-sided Kullback–Leibler divergence, while their squared Hellinger distance is quadratic in the translation. The anchor’s Theorem 2 is not a bounded-class converse, and no comparison is made with its unrestricted class.

8 Honest scope

The two-class frontier covers all integers $n \geq 1$ and $0 \leq d \leq n$, fixed $\beta \geq 1$, fixed $p \in (0, 1)$, and $B \geq 0$. At $d = 0$, every neighborhood is empty and the TTE and both minimax risks are zero; at $B = 0$, every bounded outcome is zero. If $\beta > d$, only orders through $\bar{\beta}_d = \min\{\beta, d\}$ exist. Constants are not uniform as $p \rightarrow 0$ or $p \rightarrow 1$, and can deteriorate as a nonzero $\Delta_r(p)$ approaches zero. At $p = 1/2$, a vanishing even-order contrast means only that the centered order is absent from this TTE representer, not that the corresponding coefficient is unobservable. The paper tightens the stated worst-case variance bound of [1], not SNIPE’s actual variance, and it does not claim to beat that paper’s Theorem 2 on its unrestricted class. The exact $\Psi_{b,t}$ characterization and block-local minimax constant remain confined to the ℓ_1 class on fixed complete directed blocks. Future work includes the exact leading constant for the all-measurable-estimator minimax risk and a matched frontier under heterogeneous propensities $p_j \in [p, 1 - p]$: there $\Delta_S = \prod_{j \in S} (1 - p_j) - (-1)^{|S|} \prod_{j \in S} p_j$ may vanish for some subsets of one order but not others, so no scalar exposed order $k_\star$ generally exists; the upper bound extends, but a matched design-specific converse requires new packing work. Variance estimation, studentization, and limit-law theory are outside the present theorem scope.

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